# Scaling Multi-Task Learning Beyond Homogeneous Tasks: Loss Dynamics and Gate Selection in 13-Task Financial Recommendation

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**Abstract.** Multi-task learning (MTL) architectures such as MMoE and PLE have been validated almost exclusively on 2–4 homogeneous tasks (e.g., CTR + CVR). We report empirical findings from scaling PLE to **13 heterogeneous tasks** — 7 binary, 3 multiclass, 3 regression — in a financial product recommendation system with 7 structurally distinct experts and 1M synthetic customers. Six findings challenge conventional MTL wisdom: (1) a subtle implementation bug in Kendall et al.'s uncertainty weighting — omitting per-task loss weights — silently suppresses minority-type tasks, producing a +0.018 NDCG@3 gain when fixed; (2) softmax gating outperforms sigmoid for heterogeneous task mixes, **reversing** the sigmoid advantage reported in homogeneous settings; (3) learned uncertainty weights converge to **identical** values regardless of architecture (shared-bottom vs. PLE), revealing that uncertainty weighting acts as loss-scale normalization rather than structural protection; (4) 10-epoch budgets may be insufficient for complex structures to differentiate from simpler baselines; (5) Group-TaskExpert (GTE) pre-gating **degrades** multiclass performance when groups contain mixed task types; and (6) gate entropy analysis reveals that CGC extraction-layer gating specializes meaningfully (entropy ratios 0.33–0.88 across tasks) while attention-level aggregation remains uniformly diffuse (entropy ratio 1.00), and composite val-loss is an **unreliable** checkpoint signal because regression tasks continue improving after classification/ranking metrics peak. We distill these observations into practical guidelines for practitioners scaling MTL beyond the homogeneous-task regime.

**Keywords:** Multi-task learning, Progressive Layered Extraction, Mixture of Experts, gate selection, uncertainty weighting, loss dynamics, heterogeneous tasks

## 1 Introduction

Multi-task learning (MTL) promises parameter efficiency and positive transfer by jointly optimizing related tasks [1]. In recommendation systems, MTL has become standard practice: MMoE [2] introduced multi-gate mixture-of-experts to handle task conflicts, PLE [3] added progressive extraction layers with shared and task-specific expert separation, and AdaTT [4] enabled adaptive inter-task transfer.

However, virtually all published MTL architectures for recommendation are validated on **2–4 homogeneous tasks** — typically click-through rate (CTR) and conversion rate (CVR), both binary classification problems with aligned gradient directions. When we attempted to scale PLE to **13 heterogeneous tasks** — 7 binary, 3 multiclass (4 to 50 classes), and 3 regression — several assumptions broke down in ways that existing literature does not address.

The 13-task configuration was not a design preference but a constraint. Financial regulations mandate distinct prediction targets: suitability assessment, fairness monitoring across protected attributes, churn early-warning, and product-level acquisition propensity each require a separate supervised signal. Meanwhile, limited infrastructure — a single desktop GPU (12GB VRAM) and a 3-person team — precludes maintaining separate models per task. The result is a regime that large-scale CTR teams have no reason to enter (they can afford model-per-task) but that resource-constrained regulated industries are forced into.

This paper reports six empirical findings from this scaling experience. We make no claims of state-of-the-art performance; instead, we document **phenomena and practical guidelines** that emerge when MTL is pushed beyond the homogeneous-task regime. Our contributions:

- A diagnosis of how Kendall et al.'s uncertainty weighting silently fails when per-task loss weights are omitted (Section 4.1).
- Evidence that **gate type selection depends on task-type homogeneity**, not on architectural sophistication (Section 4.2).
- A demonstration that uncertainty weights converge identically across architectures, limiting their protective role (Section 4.3).

- Analysis of epoch budget sensitivity in structure comparison (Section 4.4).
- A cautionary finding on pre-gating task grouping (GTE) with mixed-type groups (Section 4.5).
- Gate entropy analysis showing that CGC extraction-layer specialization is real and task-dependent, while attention-level aggregation collapses to uniform averaging; and a demonstration that val-loss is a misleading checkpoint criterion when regression and classification tasks coexist (Section 4.6).
- A 9-way comparison of fusion augmentations on top of the CGC baseline that identifies **two positive recipes on disjoint axes** and shows that they are **not additive**. Five representation-additive fusions (loss-level adaTT, AdaTT-sp, complementary-gate recovery, uncertainty-conditioned expert bank, and MV BRP) all degrade AUC monotonically with intervention invasiveness. **Output-space boosting with shared-expert gradient isolation** (BRP-detached) ties CGC on aggregate AUC ($\Delta = -0.0007$; best epoch exceeds baseline by $+0.0008$) while lifting F1 macro $+0.007$ and NDCG@3 $+0.015$ and retaining $+256\%$ relative rescue on the hardest multiclass task. **Training-time load-balancing regularisation** (NEAS — auxiliary supervision on the inverse-gate aggregation) is the first mechanism of the family to actually raise aggregate AUC ($\Delta = +0.0011$), with a monotone-increasing trajectory and near- uniform per-task lifts. Stacking the two positive recipes collapses NEAS's AUC gain because the shared experts cannot simultaneously be generalists (NEAS) and primary-supporting specialists (BRP-detached). The guidance is per-objective: NEAS for aggregate AUC and cross-task robustness, BRP-detached for hard-task rescue, do not stack (Section 4.7).
- A structural diagnostic showing that the causal expert's learnable adjacency matrix $\boldsymbol{W}$ collapsed to zero across every trained checkpoint we examined (four local, two upstream on-prem), rendering the expert's forward pass equivalent to a plain MLP in spite of its NOTEARS regularisation. The failure is a saddle-point problem (both task-loss and reconstruction gradients have a $\boldsymbol{W}$ factor that vanishes at the initialisation scale) and resolves under a two-part patch: add the original-paper reconstruction term ($\|\boldsymbol{z} - \boldsymbol{z}\boldsymbol{W}^2\|_F^2$) and rescale initialisation from 0.01 to 0.1. Post-patch the expert learns a valid sparse DAG ($\boldsymbol{W}$ Frobenius 0.34, 7.3% edges active, $h(\boldsymbol{W}) = 0$), but aggregate task metrics are unchanged — the DAG exists structurally but is not routed into prediction by the current architecture (Section 4.8).
- Two Axis-3 candidates that **route** the now-functional DAG into consumable outputs:

  Causal Explainability Head (CEH), a small MLP mapping the causal expert's output to a per-sample per-feature attribution vector trained via MSE against gradient $\times$ input. MV result preserves the primary AUC within noise ($+0.0015$ over the post-patch softmax baseline), marginally strengthens the DAG itself ($\boldsymbol{W}$ Frobenius $0.338 \rightarrow 0.366$, sparse edges $7.3\% \rightarrow 8.5\%$), and produces a live attribution vector for downstream audit-log persistence. A post-hoc quality evaluation revealed that under the raw grad $\times$ input target the head collapsed to a near-global importance vector (between/within-sample variance ratio 0.055, top-10 feature overlap 0.791 across samples). A minimum-intervention iteration (v2, "demeaned target") subtracts the batch mean from the supervision signal and restores per-sample discrimination without disturbing the primary task (variance ratio $0.055 \rightarrow 0.719$, top-10 overlap $0.791 \rightarrow 0.281$, primary AUC unchanged within noise, Section 4.9.4). Audit-log integration and cross-dataset reproduction remain deferred.

  Causal Guardrail (CG), a per-prediction reliability flag derived from the causal expert's latent $\boldsymbol{z}$. A W-reconstruction formulation (CG v1) fails at chance-level discrimination because the learned $\boldsymbol{W}$ is too small; a z-space Mahalanobis formulation (CG v2) detects three types of synthetic OOD at 100% TPR with 5% FPR (Section 4.10). A follow-up W-amplification experiment (W init $0.1 \rightarrow 0.3$, $\lambda_{\text{recon}} 0.5 \rightarrow 2.0$) grows the adjacency matrix $14\times$ in Frobenius norm and from 8.5% to 59.5% active edges with zero primary-task cost, partially fixing CG v1 but leaving latent-space CG v2 dominant (Section 4.11). Finding 11′s settled result: the "decorative DAG" from Finding 8 is a training-choice artefact, not an architectural constraint.

The system, data generator, and ablation scripts are publicly available.[1]

# 2 Related Work

## 2.1 Multi-Task Learning in Recommendations

Shared-bottom networks [1] suffer from negative transfer when tasks conflict. MMoE [2] mitigates this with per-task gating over a shared expert pool, and PLE [3] further separates shared and task-specific extraction

[1] https://github.com/bluethestyle/aws_ple_for_financial

layers. AdaTT [4] adds adaptive task-to-task transfer strength.

A common thread: all evaluations use 2–4 tasks of the **same type**. MMoE's Census experiment uses 2 binary tasks. PLE's production evaluation at Tencent uses 2 tasks (CTR + VCR). AdaTT's Alibaba experiment uses 3 closely related engagement tasks. No published PLE/MMoE study evaluates on a mix of binary, multiclass, and regression tasks at the scale we report.

### 2.2 Loss Balancing in MTL

Kendall et al. [5] introduced homoscedastic uncertainty weighting, learning per-task precision parameters that automatically balance loss scales. GradNorm [6] dynamically adjusts loss weights based on gradient norms. MGDA [7] solves a multi-objective optimization problem at each step.

These methods are designed for and tested on scenarios where all tasks share similar loss magnitudes. When binary cross-entropy (scale 0.5), multiclass cross-entropy (scale 3.9 for 50-class), and regression MSE (scale 0.01–1.0) coexist, the implicit assumptions of these balancing methods deserve re-examination.

### 2.3 Gate Design: Softmax vs. Sigmoid

Standard PLE and MMoE use **softmax** gates, enforcing competitive, sum-to-one expert selection. Nguyen et al. [8] demonstrated that sigmoid gating — allowing each expert to contribute independently without competition — achieves higher sample efficiency by eliminating inter-expert competition.

This finding, however, was established on homogeneous task sets. We show that the sigmoid advantage **reverses** when tasks are heterogeneous, because independent expert activation allows high-gradient binary tasks to contaminate experts that multiclass tasks depend on.

## 3 Architecture

### 3.1 PLE with Heterogeneous Expert Basket

Our PLE implementation follows Tang et al. [3] with a key modification: instead of $K$ identical MLP experts, we employ 7 **architecturally distinct** experts, each encoding a different inductive bias:

| Expert | Architecture | Input Dim |
|---|---|---|
| DeepFM | Factorization Machine + DNN | 168D |
| Temporal | Mamba + LNN + Transformer | 139D |
| HGCN | Hyperbolic GCN (Poincaré) | 27D |
| PersLay | Topological (TDA) | 32D |
| Causal | NOTEARS DAG | 161D |
| LightGCN | Graph Convolution | 100D |
| Optimal Transport | Sinkhorn matching | 127D |

Table 1: Expert basket. Each expert receives a different feature subset via FeatureRouter. Total input space: 350D (Phase 0 v3/v4).

A **FeatureRouter** assigns each expert its designated feature groups, declared in YAML configuration rather than hardcoded. The sum of per-expert input dimensions (703D) exceeds the total feature space (350D) because several feature groups are shared across experts with complementary inductive biases.

### 3.2 CGC Gating

The Customized Gate Control (CGC) module computes an attention over the $K$ experts for each task $t$:

$$g_t = \mathrm{softmax}(W_t \cdot h_t) \in \mathbb{R}^K \tag{1}$$

or

$$g_t = \sigma(w_t \cdot h_t) / \sum_j \sigma(w_j \cdot h_j) \tag{2}$$

where $h_t$ is the shared representation. **Softmax** gates enforce competitive, sum-to-one allocation. **Sigmoid** gates allow independent evaluation, normalized post-hoc. The choice between these two is a central finding of this paper (Section 4.2).

### 3.3 Uncertainty Weighting

Following Kendall et al. [5], we learn a log-variance parameter $\log \sigma_t^2$ per task. The weighted loss becomes:

$$\mathcal{L}_t^{\mathrm{uw}} = w_t \cdot \left( \frac{1}{2\sigma_t^2} \cdot \mathcal{L}_t + \frac{1}{2} \log \sigma_t^2 \right) \tag{3}$$

where $w_t$ is the per-task loss weight from configuration. The precision $\frac{1}{2\sigma_t^2}$ and the regularizer $\frac{1}{2} \log \sigma_t^2$ are clamped to stable ranges: $\log \sigma_t^2 \in [-4, 4]$, precision $\in [10^{-3}, 100]$.

The critical detail is that $w_t$ **must multiply the entire expression**, not just $\mathcal{L}_t$. Omitting $w_t$ is the bug we report in Section 4.1.

### 3.4 AdaTT Task Groups

Tasks are organized into 4 Financial DNA groups:

| Group | Tasks |
|---|---|
| Engagement | has_nba, churn_signal, next_mcc, mcc_trend, top_mcc_shift |
| Lifecycle | nba_primary, segment |
| Value | cross_sell_count, product_stability |
| Consumption | will_acquire_* (5 tasks), mcc_diversity_trend |

Table 2: Financial DNA task grouping for AdaTT transfer.

AdaTT [4] learns intra-group and inter-group transfer strengths. Note that the Consumption group mixes 5 binary tasks with 1 regression task — a design that becomes relevant in Section 4.5.

## 4 Results and Analysis

All experiments use 1M synthetic customers, 349 features (Phase 0, benchmark_v12), 10 epochs, batch size 5632, learning rate 0.0005, AMP (FP16), cosine annealing with warm restarts ($T_0 = 10$). Uncertainty weighting is applied in all runs unless noted otherwise. Each configuration is run with 3 seeds; we report medians. Metrics are reported per task type: Avg AUC (binary), Avg F1-macro (multiclass), Avg MAE (regression), NDCG@3 and Acc@3 (ranking).

### 4.1 Finding 1: The Silent Uncertainty Weighting Bug

When porting Kendall et al.'s uncertainty weighting from our on-premises codebase to the AWS implementation, a subtle bug was introduced. The on-premises code correctly implements Equation 3:

```
# On-premises (correct)
loss = loss_weight * (precision * task_loss +
log_var)
```

The AWS port omitted the per-task `loss_weight` and the clamping:

```
# AWS port (buggy)
loss = precision * task_loss + log_var / 2
```

This omission has two consequences: (1) the `loss_weight` parameter from `pipeline.yaml` — which compensates for cross-entropy scale differences between binary ( 0.5) and 50-class multiclass ( 3.9) tasks — is silently ignored; (2) without clamping, extreme log-variance values can push precision to numerically unstable regions.

The effect is that multiclass tasks, whose raw loss is 8× larger than binary tasks, receive proportionally **less** gradient when precision parameters converge to similar values. Binary tasks, being numerically dominant (7 of 13), further suppress multiclass learning through sheer gradient volume.

| Metric | Buggy | Fixed | Δ |
|---|---|---|---|
| Avg AUC (binary) | {{buggy_auc}} | {{fixed_auc}} | {{delta_auc}} |
| Avg F1 macro (multiclass) | {{buggy_f1}} | {{fixed_f1}} | +0.031 |
| NDCG@3 | {{buggy_ndcg}} | {{fixed_ndcg}} | +0.018 |
| Avg MAE (regression) | {{buggy_mae}} | {{fixed_mae}} | {{delta_mae}} |
| Val Loss | {{buggy_valloss}} | {{fixed_valloss}} | {{delta_valloss}} |

Table 3: Impact of uncertainty weighting bug fix (shared-bottom baseline). F1-macro and NDCG@3 improve substantially; binary AUC change is minimal because binary tasks were not suppressed.

**Lesson**: Uncertainty weighting implementations must be verified against the **original** mathematical formulation, not just tested for convergence. The buggy version still converges — it simply converges to a suboptimal loss balance that favors the majority task type.

### 4.2 Finding 2: Gate Selection Depends on Task Homogeneity

Existing literature suggests that sigmoid gating outperforms softmax in PLE/MoE architectures [8]. Our results show this holds only when tasks are homogeneous. With 13 heterogeneous tasks, softmax **consistently** outperforms sigmoid on ranking metrics:

| Variant | Val Loss | Avg AUC | Avg F1m | Avg MAE | NDCG@3 |
|---|---|---|---|---|---|
| Shared Bottom | {{shared_valloss}} | {{sb_auc}} | {{sb_f1m}} | {{sb_mae}} | {{sb_ndcg}} |
| PLE Softmax | {{soft_valloss}} | {{soft_auc}} | {{soft_f1m}} | {{soft_mae}} | **{{soft_ndcg}}** |
| PLE Sigmoid | {{sig_valloss}} | {{sig_auc}} | {{sig_f1m}} | {{sig_mae}} | {{sig_ndcg}} |
| PLE Sigmoid + adaTT | {{sigadatt_valloss}} | {{sigadatt_auc}} | {{sigadatt_f1m}} | {{sigadatt_mae}} | {{sigadatt_ndcg}} |

Table 4: Gate type comparison on 13 heterogeneous tasks. Softmax achieves the best NDCG@3. Bold = best per column.

**Why softmax wins here**: In a softmax gate, each task's attention over experts sums to 1, creating **competitive allocation**. When a binary task claims Expert A with weight 0.6, only 0.4 remains for other experts. This competition **isolates** experts per task type, preventing high-gradient binary tasks from contaminating experts that multiclass tasks depend on.

Sigmoid gates allow each expert to contribute independently. This is beneficial when all tasks have similar gradient magnitudes (homogeneous case), because it enables richer expert combinations. But with 7 binary tasks producing high-gradient signals and 3 multiclass tasks producing relatively smaller gradients, sigmoid allows binary gradients to flow through **all** experts simultaneously, overwhelming the weaker multiclass signal.

The decision criterion is not architectural sophistication but **task-type homogeneity**:

- Homogeneous tasks (all binary, or all regression): prefer sigmoid (richer mixing).
- Heterogeneous tasks (mixed types): prefer softmax (gradient isolation).

## 4.3 Finding 3: Uncertainty Weights Converge Identically Across Architectures

A surprising observation: the learned uncertainty weights at epoch 10 are **virtually identical** between shared-bottom and PLE-softmax:

| Task | Shared-Bottom | PLE Softmax | Δ |
|---|---|---|---|
| has_nba (binary) | {{sb_hasnba_uw}} | {{soft_hasnba_uw}} | {{delta_hasnba_uw}} |
| nba_primary (7-class) | 0.3353 | 0.3354 | +0.0001 |
| next_mcc (50-class) | 0.3360 | 0.3361 | +0.0001 |
| cross_sell (regression) | {{sb_cross_uw}} | {{soft_cross_uw}} | {{delta_cross_uw}} |
| churn_signal (binary) | {{sb_churn_uw}} | {{soft_churn_uw}} | {{delta_churn_uw}} |

Table 5: Learned uncertainty weights at epoch 10 (selected tasks). Differences are at the 4th decimal place regardless of architecture.

This means uncertainty weighting performs **loss-scale normalization** — mapping each task's loss to a comparable magnitude — but does not provide **structural protection** against gradient interference. The structural protection comes from gating (Section 4.2): with identical uncertainty weights, softmax still achieves NDCG@3 {{soft_ndcg}} vs. shared-bottom {{sb_ndcg}}, a +{{delta_ndcg_gate}} improvement attributable purely to gate structure.

**Implication**: Practitioners should not rely on uncertainty weighting alone to handle task-type conflicts. It balances scales but does not prevent gradient contamination. Gate design is the actual mechanism for protecting minority task types.

## 4.4 Finding 4: Epoch Budget Sensitivity

At 10 epochs, the performance gap between structures is small (AUC varies by ±0.002 across variants). This raises the question: are complex structures genuinely no better, or simply underfitted?

| Epoch | Shared-Bottom | PLE Softmax | PLE Sig + adaTT |
|---|---|---|---|
| 5 | {{sb_e5_auc}} | {{soft_e5_auc}} | {{sigadatt_e5_auc}} |
| 6 (NDCG peak) | {{sb_e6_auc}} | {{soft_e6_auc}} | {{sigadatt_e6_auc}} |
| 10 | {{sb_e10_auc}} | {{soft_e10_auc}} | {{sigadatt_e10_auc}} |
| 20 | {{sb_e20_auc}} | {{soft_e20_auc}} | {{sigadatt_e20_auc}} |

Table 6: Avg AUC by epoch. Complex structures may differentiate with longer training budgets. 20-epoch results pending.

NDCG@3 peaks at epoch 6, coinciding with the cosine learning rate valley before restart. This suggests that

ranking quality is sensitive to learning rate scheduling, and the apparent performance plateau at epoch 10 may reflect a scheduling artifact rather than a convergence plateau.

With cosine warm restarts ($T_{\text{mult}} = 2$), the second cycle spans epochs 10–30. Complex structures (PLE + adaTT) have more parameters to warm up, and may only differentiate from simpler baselines in the second cosine cycle.

**Guideline**: When comparing MTL structures, ensure the training budget is at least $2 \times T_0$ to observe post-restart behavior. 10-epoch comparisons may prematurely favor simpler architectures.

## 4.5 Finding 5: GTE Pre-Gating Degrades Mixed-Type Groups

GroupTaskExpert (GTE) adds a pre-gating layer that partitions expert representations by task group **before** PLE gating occurs. The motivation is to strengthen intra-group expert sharing.

However, when a task group contains mixed types — the Consumption group has 5 binary tasks and 1 regression task — GTE forces shared representation learning across incompatible loss types:

| Variant | NDCG@3 | Avg AUC |
|---|---|---|
| PLE Softmax | {{soft_ndcg}} | {{soft_auc}} |
| PLE Sigmoid + GTE | {{siggte_ndcg}} | {{siggte_auc}} |
| Δ | {{delta_gte_ndcg}} | {{delta_gte_auc}} |

Table 7: GTE impact. NDCG@3 drops substantially when GTE forces mixed-type groups to share pre-gate representations.

The mechanism: GTE pools expert outputs at the group level **before** per-task PLE gating can differentiate them. Within the Consumption group, the 5 binary tasks' high-gradient signals dominate the pooled representation, and the single regression task (`mcc_diversity_trend`) receives a representation already biased toward binary decision boundaries. PLE gating, which operates **after** GTE pooling, cannot undo this damage.

**Guideline**: Task groups for GTE (or similar pre-gating mechanisms) must be homogeneous by task type, not by business semantics. If a business-meaningful grouping mixes types, omit GTE and rely on PLE gating alone for inter-expert allocation.

## 4.6 Finding 6: Gate Entropy and Loss–Metric Decoupling

### 4.6.1 CGC Gate Entropy Analysis

To understand **how** PLE's CGC gating allocates experts in practice, we compute the Shannon entropy ratio of each task's gate weight distribution $g_t \in \mathbb{R}^K$ at the end of teacher training (30 epochs). The entropy ratio is defined as:

$$E_t = \frac{H(g_t)}{H_{\max}} = -\frac{\sum_k g_{\{t,k\}} \log g_{\{t,k\}}}{\log} K \tag{4}$$

where $H_{\max} = \log K$ is the maximum entropy for $K$ experts. $E_t = 1$ means perfectly uniform expert utilization; $E_t = 0$ means a single expert captures all weight.

| Task | Layer 1 | Layer 2 | Pattern |
|---|---|---|---|
| top_mcc_shift | 0.347 | — | Single-expert dominance |
| product_stability | 0.431 | — | Single-expert dominance |
| segment_prediction | — | 0.332 | Single-expert dominance |
| cross_sell_count | 0.570 | 0.614 | Moderate diversity |
| churn_signal | 0.691 | 0.860 | Moderate → full diversity |
| nba_primary | 0.851 | 0.839 | Full expert utilization |
| will_acquire_payments | 0.882 | — | Full expert utilization |

Table 8: CGC gate entropy ratios by task and PLE layer (teacher model, 30 epochs). Low entropy ($E_t < 0.45$) indicates 1–2 experts dominate; high entropy ($E_t > 0.80$) indicates all 7 experts contribute meaningfully.

The entropy ratios reveal three behaviorally distinct task clusters:

**Single-expert dominance** ($E_t$ 0.33–0.43): Tasks such as `top_mcc_shift` (MCC category shift prediction) and `segment_prediction` (3-class customer segment) are captured by 1–2 experts. These tasks appear to encode simple patterns that a single specialized expert — DeepFM for transactional features, or HGCN for hierarchical segments — handles near-optimally. The low entropy is not a failure mode; it is efficient routing.

**Moderate diversity** ($E_t$ 0.57–0.72): Tasks such as `cross_sell_count` (count regression) and some binary acquisition tasks draw on 3–4 experts. These tasks likely require both transactional signals (DeepFM) and

sequence-level patterns (Temporal), explaining partial expert spread.

**Full expert utilization** ($E_t$ 0.85–0.88): `nba_primary` (7-class next best action) and `will_acquire_payments` (binary) actively use all 7 experts. These tasks encode complex, multi-faceted customer behavior patterns that no single architectural inductive bias fully captures.

#### 4.6.2 Attention Collapse: A Structural Blind Spot

At the attention aggregation level (shared expert aggregation), all 13 tasks exhibit an entropy ratio of **exactly 1.000**. The attention mechanism has not learned to differentiate — it acts as a simple average over the expert pool.

This is structurally significant. The CGC extraction-layer gates (Table Table 8) demonstrate that **the model can learn differentiated expert preferences**, but this capacity is entirely absent at the attention-aggregation level. Two possible explanations:

1. **Gradient starvation**: The attention parameters receive gradients only after passing through per-task tower heads, which already specialize via the extraction-layer gates. By the time signal reaches the attention layer, per-task distinction may be adequately handled upstream.
2. **Parameterization bottleneck**: If the attention query dimension is small relative to the expert embedding space, the attention has insufficient capacity to form task-specific preferences across 7 heterogeneous experts.

In either case, the attention component adds parameters without performing meaningful routing. This is a candidate for architectural simplification — replacing attention aggregation with a fixed average or a learned scalar per task — in future work.

#### 4.6.3 Loss–Metric Decoupling at 30 Epochs

Extending training from 10 to 30 epochs (with $T_0 = 10$, cosine warm restarts) exposes a fundamental tension in composite loss monitoring.

| Epoch | Val Loss | Avg AUC | NDCG@3 | Avg MAE |
|---|---|---|---|---|
| 1 | 32.11 | — | — | 1.07 |
| 10 | 26.43 | **0.6726** | **0.6976** | 1.01 |
| 11 | 25.89 | 0.6702 | 0.6853 | 0.99 |
| 19 | 24.01 | 0.6718 | 0.7004 | 0.97 |
| 20 | 23.52 | 0.6691 | 0.6657 | 0.96 |
| 29 | 22.95 | 0.6704 | 0.6540 | 0.96 |
| 30 | 22.68 | 0.6687 | — | 0.96 |

Table 9: Loss–metric decoupling over 30 epochs (teacher model, 1M customers, $T_0 = 10$ cosine warm restarts). Val loss decreases monotonically while AUC peaks at epoch 10 (−0.4%p by epoch 30) and NDCG@3 peaks at epoch 19 then collapses (−4.4%p by epoch 29). Avg MAE continues improving throughout. Bold = metric peak across all epochs.

Val loss decreases monotonically (32.11 → 22.68), which would conventionally indicate continuous improvement. However:

- **Avg AUC** (binary tasks): peaks at epoch 10 (0.6726), then declines to 0.6687 by epoch 30 (−0.4%p). The decline is modest but consistent.
- **NDCG@3** (ranking quality): peaks at epoch 19 (0.7004), then collapses to 0.6540 at epoch 29 (−4.6%p relative to peak, −3.0%p relative to epoch 10).
- **Avg MAE** (regression tasks): improves steadily, 1.07 → 0.96, throughout all 30 epochs.

The root cause is **task-type dominance in composite loss**: regression tasks contribute continuously shrinking MAE to the composite loss and pull the aggregate downward even as classification tasks saturate. With 3 regression tasks whose losses have no natural lower bound (unlike cross-entropy, which approaches 0 for well-separated data), the composite loss signal is **not a valid proxy** for classification or ranking quality after the first cosine cycle.

#### 4.6.4 Cosine Restart Oscillation Across Task Types

Cosine warm restarts ($T_0 = 10$) create learning rate spikes at cycle boundaries. NDCG@3 exhibits strong oscillation at these boundaries:

| Epoch (event) | NDCG@3 | Change |
|---|---|---|
| 10 (cycle 1 end) | 0.6976 | peak |
| 11 (restart 1) | 0.6853 | −1.2%p (sharp drop) |
| 19 (cycle 2 near-end) | 0.7004 | recovery, new peak |
| 20 (restart 2) | 0.6657 | −3.5%p (sharp drop) |
| 29 (cycle 3 near-end) | 0.6540 | recovery failure |

Table 10: NDCG@3 oscillation at cosine restart boundaries. Each LR spike pushes the model away from the ranking-optimal parameter region. The second restart produces a larger drop than the first, and the third cycle fails to recover, indicating progressive divergence.

The pattern is asymmetric across task types. Avg MAE is **unaffected** by restarts — regression loss landscapes are smooth and the optimizer quickly returns to a low-MAE region after each LR spike. Binary AUC shows a small dip (−0.4%p per restart) and recovers partially. NDCG@3 suffers the largest disruption (up to −3.5%p per restart) because ranking metrics are sensitive to the relative ordering of scores, and LR restarts temporarily scramble the score scale before the model re-converges.

**Implication**: For ranking-sensitive applications, cosine restarts with $T_{\text{mult}} = 1$ (constant cycle length) should be replaced with $T_{\text{mult}} = 2$ (doubling cycle length) or with learning rate warmup-then-decay (no restart), evaluated over a 30-epoch budget.

#### 4.6.5 Checkpoint Selection Criterion

These findings jointly establish that **val loss is an invalid checkpoint criterion** when tasks of different types share a composite loss. The correct approach is a composite checkpoint metric that weights metric semantics by task type:

$$\text{score}_{\text{ckpt}} = \alpha \cdot \text{AvgAUC} + \beta \cdot \text{NDCG@3} + \gamma \cdot (1 - \text{NormMAE}) \tag{5}$$

where $\alpha, \beta, \gamma$ are set to weight task types equally (e.g., $\alpha = \beta = \gamma = \frac{1}{3}$) rather than proportional to task count. In our 13-task configuration, using this composite score selects epoch 10 as the optimal checkpoint, matching the AUC peak — not epoch 29, which minimizes val loss.

**Guideline**: When regression tasks are present in a heterogeneous MTL system, (1) define a composite checkpoint metric across task types before training begins, (2) checkpoint every epoch and select post-hoc rather than monitoring val loss, and (3) treat val loss as a diagnostic (indicating regression progress) rather than as the primary stopping criterion.

### 4.7 Finding 7: Two Positive Fusion Recipes on Disjoint Axes, Non-Additive Under Composition

After Finding 2 established that the loss-level `adaTT` variant does not affect aggregate AUC at the 13-task scale ($\Delta = -0.001$, null), eight further mechanisms were evaluated on the same benchmark to map out which forms of fusion augmentation (if any) can extract useful signal beyond CGC's gated selection. The nine-way comparison resolves into three regions. **Representation-additive fusions** (five variants: loss- level adaTT, AdaTT-sp, M1 complement, ECEB, and BRP-MV) all inject a residual into the primary representation or propagate residual-error gradients into shared experts; all five degrade aggregate AUC with a magnitude that scales monotonically with the invasiveness of the intervention. **Output-space boosting with gradient isolation** (BRP- detached) preserves AUC ($\Delta = -0.0007$, tied) while lifting F1 macro by +0.007 and NDCG@3 by +0.015. **Training-time load-balancing regularisation** (NEAS — an inverse-gate auxiliary supervision) is the first mechanism of the family to **raise** aggregate AUC ($\Delta = +0.0011$), with near-uniform small lifts across most tasks. The two positive recipes act on disjoint axes — error correction in output space vs. expert-collapse prevention at the gate — yet a ninth experiment stacking them produces a **non-additive** outcome: NEAS's AUC lift vanishes while BRP-detached's hard-task rescue partially survives, because the shared experts cannot simultaneously generalise for NEAS and specialise for the primary-supporting optimum that BRP-detached relies on.

#### 4.7.1 Five Mechanisms, One Aggregate-AUC Conclusion

**Loss-level adaTT** (the variant reported in Paper 1) adds a weighted cross-task loss term, $L_i + \lambda \sum_{j \neq i} w(i \to j) L_j$, with transfer weights estimated from gradient cosine similarity.

**AdaTT-sp** [4] adds a native-expert residual: after the CGC gate produces the weighted sum, the mean output of the task's own task-specific experts is re-injected, scaled by a learnable scalar.

**Residual complement (M1)**, introduced in this paper, preserves the primary gated output and adds a complementary weighting $(1 - \text{gate_weights})$ (clamped and renormalised over the expert axis) applied to the same expert outputs as a residual, scaled by a learnable scalar. The intent is to recover intra-task signal from

experts the gate down-weighted, without any cross-task mixing.

**ECEB (Error-Conditioned Expert Bank, MV)**, introduced in this paper, is designed specifically to escape the shared structure of the three above: the residual is derived from the gate's **entropy** rather than from the gate's output. Concretely, the recovery path is a task-agnostic consensus (mean over all expert outputs, no gate weighting), scaled at forward time by the product of a per-task learnable scalar $\sigma(w_t)$ and the normalised gate entropy $\frac{H(g_t)}{\log} N$ (per sample). When the gate is confident (low entropy), recovery collapses toward zero; when it is distributed, recovery activates. By construction ECEB eliminates the "gate-derived residual" factor.

**BRP (Boosting-Residual Path, MV)**, also introduced in this paper, removes the additive-on-representation structure altogether. A per-task residual expert bank takes the last CGC layer's `shared_concat` (gate-bypass feature view) and produces a logit residual matching the primary tower's output shape. The residual is trained by MSE against the primary's **detached** prediction error, i.e. $y - \text{activation}(\text{stop_grad}(\text{primary}))$ (single-stage boosting). The primary pathway trains on ground truth alone; the combined output $\text{primary} + \sigma(\lambda_t) \cdot \text{residual}$ is used for inference and evaluation only. The primary representation is never touched.

**BRP-detached**, a single-line modification of BRP, feeds `shared_concat.detach()` into the residual bank instead of the raw `shared_concat`. This cuts any residual-MSE gradient from flowing back into the shared experts while leaving the primary pathway, parameter count, and training schedule unchanged. The modification was motivated by the per-task analysis of BRP below.

**NEAS (Neglected-Expert Auxiliary Supervision)** adds a per-task auxiliary head consuming the **inverse-gate-weighted aggregation** of the last CGC layer's expert outputs. The auxiliary target is the primary task label, and the auxiliary loss (scaled by `aux_weight = 0.05`) is added to the total loss during training only; inference does not use the auxiliary head at all. The mechanism explicitly forces neglected experts — those the gate de-emphasises for a given task — to retain predictive representations, mitigating expert collapse. NEAS is structurally independent of all residual mechanisms above; it neither injects a residual nor modifies the primary output.

**NEAS + BRP-detached** stacks the two positive mechanisms, testing additivity. Both are enabled with their standalone settings; no other modification.

Results on the 13-task benchmark (10 epochs, seed=42):

| Fusion | Final AUC | F1 macro | NDCG@3 | Δ AUC |
|---|---|---|---|---|
| CGC gate (baseline) | 0.6728 | 0.2002 | 0.6820 | — |
| Loss-level adaTT | 0.6717 | 0.2013 | 0.6646 | −0.0011 |
| AdaTT-sp (Li 2023) | 0.6696 | 0.1998 | 0.6570 | −0.0032 |
| M1 complement | 0.6675 | 0.1998 | 0.6611 | −0.0053 |
| ECEB (MV) | 0.6665 | 0.1998 | 0.6549 | −0.0063 |
| BRP (MV) | 0.6650 | **0.2117** | **0.7039** | −0.0078 |
| BRP-detached | 0.6721 | 0.2075 | 0.6965 | −0.0007 (tied) |
| **NEAS** | **0.6739** | 0.2019 | 0.6896 | **+ 0.0011 (positive)** |
| NEAS + BRP-detached | 0.6722 | 0.2062 | 0.6864 | −0.0006 (non-additive) |

Table 11: 9-way comparison of fusion mechanisms on the 13-task benchmark. The five representation-additive variants (rows 2–6) degrade aggregate AUC below CGC, with magnitude growing monotonically in the invasiveness of the intervention. BRP-detached (row 7) ties CGC on AUC ($\Delta = -0.0007$, best $= 0.6736$ at epoch 8, $+$ 0.0008 over the baseline final) and lifts F1/NDCG@3. NEAS (row 8) is the first mechanism of the family to **raise** aggregate AUC ($\Delta = +0.0011$), with a monotone-increasing trajectory through all 10 epochs and near-uniform small per-task lifts. The combined scenario (row 9) collapses NEAS's AUC gain — the two mechanisms exert opposing pressures on the shared experts and do not stack.

M1′s best AUC at epoch 1 (pre-training) with monotone decline thereafter indicates that training the learnable recovery weight actively degrades performance — random initialisation is a less harmful operating point than the converged weight.

#### 4.7.2 Per-Task Breakdown and the Two Outliers

Aggregate deltas are at noise level ($\leq 0.005$) across all variants, but a per-task breakdown reveals three regimes:

- **Gate-saturated tasks** (segment_prediction, top_mcc_shift, mcc_diversity_trend) have low gate entropy (ratio $< 0.55$) and are insensitive to every recovery mechanism ($|\Delta| \leq 0.003$).
- **Gate-distributed tasks with a strong primary** (churn_signal, will_acquire_lending) have high gate entropy (ratio $> 0.82$) and show the largest M1 degradation ($-0.020$ and $-0.009$).
- **A single positive outlier** is next_mcc (50 classes, near-random base F1 $\approx 0.01$): all three recovery variants improve it by $+0.005$ to $+0.008$. The gain is large relative to the base but small in absolute terms; we attribute it to the near-floor starting point rather than to a genuine recovery effect.

The remaining 8 tasks fall within noise ($|\Delta| \leq 0.005$).

#### 4.7.3 Gate Entropy Correlation: Weak Signal

To test whether gate entropy structurally predicts recovery benefit, we extract per-task gate weights from the final CGC layer of the joint_full checkpoint and correlate task-level entropy with each variant's $\Delta$:

- Loss-level adaTT: $r = -0.31$
- AdaTT-sp: $r = -0.32$
- M1 complement: $r = -0.40$

All three correlations are negative (higher entropy $\rightarrow$ more harm) with consistent sign, but with $n = 13$ and $p \approx 0.18$ none meets conventional significance. The two outliers — churn_signal and next_mcc — are better explained by task-specific factors (label construction for churn_signal, near-floor base rate for next_mcc) than by gate entropy. Gate entropy cannot therefore be claimed as a structural predictor of recovery benefit on this benchmark.

#### 4.7.4 BRP, BRP-detached, and NEAS: Diagnosing the Mechanisms

The first four augmented variants (loss-level adaTT, AdaTT-sp, M1, ECEB) all inject a residual additively into the primary representation and produce monotone degradation. BRP is the one variant in the family that places the residual in output space. BRP's aggregate AUC is nevertheless the **lowest** of the five non-baseline runs ($\Delta = -0.0078$), but the drop is accompanied by the **highest** F1 macro and NDCG@3 of those five ($+0.0115$ and $+0.0219$ over CGC). The BRP result therefore looked at first like a task-balance trade-off rather than a success.

A per-task breakdown tells a more specific story. next_mcc (50-class, baseline macro-F1 at near-random 0.0100) improves to 0.0440 (+340% relative) under BRP — the hard-task rescue effect. In the opposite direction, churn_signal — the task with the highest binary AUC in the baseline (0.6868) — drops to 0.6512 ($-0.036$) under BRP, and this single task accounts for most of the aggregate AUC loss. Excluding churn_signal, BRP's binary AUC sits near baseline; the remaining five binary tasks each lose between $-0.001$ and $-0.010$.

The mechanism is a shared-expert gradient leak. BRP's residual bank consumes `shared_concat`, and residual-MSE gradients propagate back into the shared experts — only the residual **target** was detached, not its input. For tasks where the primary is already near its ceiling, the shared experts had converged to a primary-supporting optimum and the additional MSE pressure pulls them off it. For tasks where the primary struggles (next_mcc), the residual supplies signal the primary could not extract. The aggregate drop is therefore not an algorithmic limitation of output-space boosting — it is an implementation artefact of training the residual bank through the shared representation.

BRP-detached tests this directly. Swapping `shared_concat` for `shared_concat.detach()` in the residual bank's input — no parameter change, no training-schedule change — produces the following per-task pattern:

| Task | CGC | BRP | BRP-detached | Verdict |
|---|---|---|---|---|
| churn_signal (AUC) | 0.6868 | **0.6512** | **0.6852** | restored |
| will_acquire_lending (AUC) | 0.6549 | 0.6453 | 0.6553 | restored |
| will_acquire_deposits (AUC) | 0.6534 | 0.6493 | 0.6536 | restored |
| will_acquire_investment (AUC) | 0.6754 | 0.6719 | 0.6764 | restored |
| next_mcc (F1 macro) | 0.0100 | **0.0440** | **0.0356** | retained (+256%) |
| remaining 8 tasks | — | ±0.002 | ±0.002 | unchanged |

Table 12: Per-task comparison of BRP and BRP-detached versus CGC on the subset of tasks where BRP materially changed the metric. Detaching `shared_concat` restores every easy-task AUC loss (−0.036 on churn_signal drops to −0.002) while retaining the majority of the next_mcc rescue effect (+340% relative reduces to +256%, still dominant over CGC's near-random baseline).

The diagnosis is therefore confirmed at the per-task level. The easy- task AUC loss in BRP was caused by residual-MSE gradients reshaping shared experts; detaching the input cuts that channel, shared experts remain on the primary-supporting optimum, and the residual bank uses only its own parameters to learn the hard-task correction.

**NEAS** takes a different path. Rather than a residual, it attaches an auxiliary supervision signal to the **inverse-gate-weighted aggregation** of expert outputs: each task's aux head must predict the primary label using mostly the experts the gate de-emphasised. This creates gradient pressure on every shared expert to remain predictive even when the task-level gate concentrates. The effect is a training- time regulariser against expert collapse. NEAS's trajectory rises monotonically through all 10 epochs, and its per-task lift is spread across 11 of 13 tasks (six of seven binaries improve by +0.0004 to + 0.0029; nba_primary lifts F1 by +0.0056). Unlike BRP, NEAS does not produce a large rescue on any single hard task — next_mcc's F1 moves only from 0.0100 to 0.0107 under NEAS alone, against +0.0256 under BRP-detached — because its mechanism is prevention-of-loss rather than targeted-correction.

The two positive recipes therefore act on **disjoint axes**:

| Dimension | BRP-detached | NEAS |
|---|---|---|
| Where it acts | Output-space residual | Shared-expert gradients via aux loss |
| Training signal | MSE on primary's detached error | Task loss on inverse-gate aggregation |
| Inference overhead | Non-zero (residual expert) | Zero (training-only regulariser) |
| Parameter addition | 0.36M (residual bank) | 0.17M (aux heads) |
| Aggregate Δ AUC | −0.0007 (tied) | +0.0011 (positive) |
| Typical per-task pattern | One big rescue (+256% next_mcc) | Many small lifts (±0.003) |
| Failure mode if stacked | Retains next_mcc rescue | AUC lift erased |

Table 13: Structural comparison of the two positive fusion recipes identified on this benchmark.

#### 4.7.5 Conclusion: Three Structural Regions, Non-Additive Composition

Across the nine runs, three structural regions organise the space of fusion augmentations on top of a heterogeneous-expert PLE with CGC gating:

1. **Representation-additive fusions fail on aggregate AUC.** Five variants — loss-level adaTT, AdaTT-sp, M1 complement, ECEB, and BRP-MV (whose `shared_concat` input propagates residual gradients into shared experts) — all inject or propagate residual-error signal into the primary-representation path. All five degrade AUC, with a monotone relationship between intervention invasiveness and degradation magnitude (−0.001 to −0.008). The residual's definition (gate inverse, own expert, uncertainty-gated consensus, boosting error) does not change this; what matters is whether the mechanism reaches into the primary-supporting representation.
2. **Output-space boosting with gradient isolation succeeds as a tie- plus-ranking-gain recipe.** BRP-detached places the residual in output space and trains it as a boosting correction on the primary's detached error, while detaching the `shared_concat` input to the residual bank so residual-MSE gradients never reach the shared experts. The result ties CGC on aggregate AUC ($\Delta = -0.0007$; best epoch exceeds baseline by +0.0008) and lifts F1 macro by +0.007

and NDCG@3 by +0.015, with +256% relative rescue on the hardest multiclass task.

3. **Training-time load-balancing regularisation succeeds as an AUC- positive recipe.** NEAS attaches a per-task auxiliary head to the inverse-gate-weighted aggregation of expert outputs, trained against the primary label. It is the first mechanism in this family to actually raise aggregate AUC ($\Delta = +0.0011$), with a monotone- increasing trajectory over 10 epochs and near-uniform per-task lifts. The mechanism is zero-overhead at inference (training-only regulariser) and adds only 0.17M parameters.

**The two positive recipes act on disjoint axes** (error correction in output space vs. expert-collapse prevention at the gate) and are independently reproducible. A ninth experiment stacking them produces a **non-additive** outcome: NEAS's aggregate AUC lift vanishes ($\Delta = -0.0006$), while BRP-detached's hard-task rescue partially survives (next_mcc F1 +0.0250 vs. +0.0256 standalone). The mechanism-level explanation is that NEAS pushes shared experts toward **generalists** (each must predict under inverse-gate reweighting), whereas BRP-detached holds shared experts on the **primary-supporting optimum** via the primary task loss. The shared experts are a finite resource and cannot simultaneously satisfy both pressures.

The practical reading is therefore per-objective rather than stack- everything:

- If aggregate AUC and uniform cross-task robustness matter most, use **NEAS** (zero inference overhead, +0.0011 AUC, monotone training trajectory).
- If rescue of a near-random multiclass task matters most, use **BRP-detached** (ties AUC, +256% relative F1 rescue on the hardest task).
- Do **not** stack them as-is; doing so erases NEAS's AUC gain without producing the hoped-for additive lift. A mechanism that resolves the shared-expert conflict (e.g., a scheduler that alternates NEAS and BRP-detached pressure across training phases, or a parameter- sharing adapter between the two heads) is a natural follow-up but not studied here.

## 4.8 Finding 8: The Causal Expert's Adjacency Matrix Was a Dead Parameter

During preparatory diagnostics for a follow-up study reinterpreting the role of the causal expert, an unexpected failure surfaced in the baseline architecture itself: the causal expert's learnable adjacency matrix $\boldsymbol{W} \in \mathbb{R}^{32\times 32}$ — the DAG structure the expert is supposed to learn — had collapsed to essentially zero across every checkpoint we inspected. The effect is general, not an artefact of any particular scenario:

| **Checkpoint** | $\boldsymbol{W}$ **Frobenius** | **Entries** $\lvert W\rvert > 0.01$ |
|---|---|---|
| struct_13_ple_sigmoid (CGC baseline) | 0.0001 | 0% |
| struct_13_residual_complement (M1) | 0.0001 | 0% |
| struct_13_eceb (MV) | 0.0003 | 0% |
| struct_13_brp_detached | 0.0001 | 0% |
| upstream on-prem implementation | 0.0001 | 0% |

Table 14: Causal expert's adjacency matrix across five independently trained checkpoints, including two from the upstream on-prem implementation that predates the port. In every case the Frobenius norm of $\boldsymbol{W}$ is below its random init scale, and not a single off-diagonal entry exceeds 0.01 in magnitude. The DAG the expert is supposed to learn does not exist after training.

### 4.8.1 Root Cause: A Residual That Bypasses the DAG

The expert's forward pass is

$$\boldsymbol{z}_{\text{hat}} = \boldsymbol{z} + \boldsymbol{z}\boldsymbol{W}^2 \tag{6}$$

where $\boldsymbol{z} = \text{feature_compressor}(\boldsymbol{x})$ and $\boldsymbol{z}_{\text{hat}}$ feeds a downstream `causal_encoder` MLP. The residual term $\boldsymbol{z}$ carries the full latent content regardless of $\boldsymbol{W}$, so the task loss has no structural incentive to push $\boldsymbol{W}$ away from zero. The NOTEARS acyclicity and sparsity regularisers both penalise $\boldsymbol{W}$ away from being dense, but neither penalises $\boldsymbol{W} = 0$ — in fact the sparsity term is **minimised** there. The global optimum of the combined objective is therefore $\boldsymbol{W} = 0$, and training converges to it reliably.

The learned gradient carries the same problem. Both the task-loss contribution via $\partial \boldsymbol{z}_{\text{hat}}/\partial \boldsymbol{W}$ and the NOTEARS reconstruction gradient (when added) are proportional to $\boldsymbol{W}$ itself:

$$\frac{\partial}{\partial \boldsymbol{W}} \operatorname{trace}\left((\boldsymbol{W} \odot \boldsymbol{W})^k\right) \propto \boldsymbol{W} \tag{7}$$

Any near-zero initialisation produces a near-zero gradient, so $\boldsymbol{W} = 0$ is a **saddle point** that the optimiser cannot escape on its own.

### 4.8.2 Patch: Reconstruction Loss + Initialisation Rescale

Two changes, both load-bearing:

1. **Reconstruction regulariser.** The original NOTEARS paper minimises $\|\boldsymbol{X} - \boldsymbol{X}\boldsymbol{W}\|_F^2$ — an explicit reconstruction signal. We adopt the compressed-latent analogue as a third term in `get_dag_regularization()`:

$$\mathcal{L}_{\text{recon}} = \text{mean}\left(\left(\boldsymbol{z} - \boldsymbol{z}\boldsymbol{W}^2\right)^2\right), \quad \lambda_{\text{recon}} = 0.5 \quad (8)$$

   This re-introduces the direct pressure on $\boldsymbol{W}$ that the original paper relies on.

2. **Initialisation rescale.** The initial $\boldsymbol{W} \sim \mathcal{N}(0, 0.01^2)$ was too small: its $\boldsymbol{W}^2$ entries sit at $10^{-4}$, a scale at which the gradient of either task or reconstruction loss is effectively zero. Rescaling the init to $\mathcal{N}(0, 0.1^2)$ keeps $\boldsymbol{W}^2$ on an $O(10^{-2})$ scale (still a small perturbation to the residual path) while carrying enough magnitude to propagate gradient during early training.

Either change alone is insufficient. Reconstruction without init rescale was verified on a 10-epoch SageMaker run and left $\boldsymbol{W} \approx 0$; init rescale without reconstruction would restore the standard NOTEARS pressure but leave the gradient vanishing problem.

#### 4.8.3 Post-Patch Verification

A 10-epoch SageMaker run with both changes produces a non-trivial DAG for the first time on this codebase:

- $\boldsymbol{W}$ Frobenius: **0.338** (init scale was 0.1)
- $|W|$-threshold sparsity at 0.01: 7.3%
- Acyclicity $h(\boldsymbol{W}) = 0$ (valid DAG)
- Mean self-loop strength: 0.000 (diagonal suppressed as intended)
- Top edges by $W_{ij}^2$: `var_23 -> var_13` (0.019), `var_9 -> var_13` (0.009), `var_15 -> var_11` (0.007)

The sparsity ratio is in the target range (paper-recommended 5–15%) and the top edges show a consistent sink (`var_13`), which is the kind of hub structure a latent DAG ought to exhibit.

#### 4.8.4 Aggregate Task Metrics Are Unchanged

Patching the collapse does **not** change downstream task performance. On the same SageMaker softmax-gate 10-epoch run, aggregate metrics are within noise of the pre-patch softmax baseline:

| **Run** | **AUC** | **F1 macro** | **NDCG@3** | **MAE** |
|---|---|---|---|---|
| Pre-patch local (softmax, $\boldsymbol{W} \approx 0$) | 0.6729 | 0.2009 | 0.6814 | 0.9598 |
| Post-patch SageMaker (softmax, $\boldsymbol{W}$ learned) | 0.6719 | 0.2042 | 0.6875 | 0.9597 |
| $\Delta$ | −0.001 | +0.003 | +0.006 | 0 |

Table 15: Task metric change after the W-collapse patch. Differences across AUC, F1 macro, NDCG@3, and MAE are within the noise band observed across the 9-way fusion comparison (Finding 7). The structural bug was real, but its resolution does not, by itself, translate into task improvement.

#### 4.8.5 Implication

The diagnostic finding is structural but the metric finding is null. Two readings:

1. The causal expert has been contributing to prediction primarily through its `causal_encoder` MLP, which is a regular non-linear transform that does not require a meaningful $\boldsymbol{W}$ to fit. Adding a meaningful $\boldsymbol{W}$ changes the latent pathway ($\boldsymbol{z}_{\text{hat}} = \boldsymbol{z} + \boldsymbol{z}\boldsymbol{W}^2$) by a small amount, but the downstream encoder adapts and the ensemble's final prediction looks the same. The DAG is, in the current architecture, **decorative** rather than functionally used by the prediction.
2. For explainability claims that depend on the DAG (attribution, counterfactual probes, reason-code generation), this matters. Pre-patch the DAG was absent — so any such claim built on `get_causal_graph()` was retrieving essentially noise at init scale. Post-patch the DAG exists and is structured, but is not itself routed into the prediction path.

This motivates a separate structural study (beyond this paper's scope) that redefines the causal expert's role so the learned DAG has a load-bearing route to prediction, not just to the expert's own internal representation. Candidates include an attribution head that consumes the DAG directly, a routing signal from the causal expert to the per-task gate, and a counterfactual probe head. The W- collapse patch reported here is a pre-condition for any of those explorations — without it, the DAG is not there to be routed.

## 4.9 Finding 9: Causal Explainability Head (CEH) — First Axis-3 Attempt

As the first of several Axis-3 candidates that re-wire the causal expert's role (CEH / CG / CTGR / CRCG / CCP), CEH adds a per-sample per-feature attribution vector on top of the causal expert. The motivation is explicit: now that Finding 8 gives the expert a real DAG, the DAG needs to leave the expert's internals and reach a consumer. CEH is the shortest path from "DAG exists" to "DAG has a per-prediction output that an audit log can persist."

### 4.9.1 Design

A small MLP head maps the expert's 64-dim output back to an input_dim-wide attribution vector, and is trained via MSE to align with the gradient × input saliency baseline of the expert's own scalar output w.r.t. the input:

$$\mathcal{L}_{\text{attr}} = \text{mean}\Big(\Big|\text{head}(\text{output}) - \Big(\nabla_x \text{ output}. \sum() \odot x\Big)\Big|^2\Big) \tag{9}$$

Gradient × input is computed with one extra forward pass on a cloned, `requires_grad=True` copy of the input, so the main forward graph is untouched. Cost is ≈ 14% extra compute inside the causal expert (about 1/7 of shared-expert time). Primary prediction path is unchanged; CEH is a training-time regulariser plus an inference-time side output.

Inference: `expert._last_attribution` exposes a per-sample vector of length input_dim. The pipeline-side integration with the audit log (HMAC-signed persistence, SR 11-7 MRM) is Paper 2 v2 scope.

### 4.9.2 MV Result (SageMaker teacher_full + CEH, 10 epochs, softmax gate)

| Metric | Pre-patch (W ≈ 0) | Post-patch (no CEH) | Post-patch + CEH |
|---|---|---|---|
| Primary AUC | 0.6729 | 0.6719 | **0.6734** |
| F1 macro | 0.2009 | 0.2042 | 0.1994 |
| NDCG@3 | 0.6814 | 0.6875 | 0.6842 |
| MAE | 0.9598 | 0.9597 | 0.9609 |
| $\boldsymbol{W}$ Frobenius | 0.0001 | 0.338 | **0.366** |
| $\lvert W\rvert > 0.01$ sparse edges | 0% | 7.3% | **8.5%** |
| Attribution head trained | n/a | n/a | **yes** |

Table 16: CEH MV result. Primary AUC is preserved within noise; $\boldsymbol{W}$ Frobenius and sparse-edge share are both slightly higher than the post-patch no-CEH run, suggesting the attribution-head gradient provides an additional structural signal that reinforces the DAG. Attribution head training was verified by post-hoc inspection of the head's weights and biases (non-zero biases starting from zero-init).

The attribution head's layer-3 bias (init = $\mathbf{0}$) moved to $\|\boldsymbol{b}\| = 0.03 \pm 0.04$ per element and layer-0 bias to $0.08 \pm 0.08$, which is straightforward evidence that the MSE alignment loss was contributing gradient to the head.

### 4.9.3 What the MV Does Not Verify

CEH MV confirms that:

1. The head trains without disrupting the primary pathway.
2. The W-collapse patch (Finding 8) remains intact under the extra attribution-training signal — in fact the DAG's Frobenius norm and sparse-edge count both increase marginally.
3. The per-prediction attribution vector exists and can be consumed downstream.

CEH MV does **not** yet verify:

- **Attribution quality.** The head fits the gradient × input target by construction; whether the learned output carries per-sample signal or merely reproduces a global importance pattern required a dedicated post-hoc evaluation. That evaluation (Section 4.9.3) showed a near-global collapse under the raw target

and motivated the v2 iteration (Section 4.9.4), which resolved the collapse.

- **Audit-log utility.** The per-sample vector is now available but has not yet been routed into an HMAC-signed, append-only audit log record that regulators can query. That integration is Paper 2 v2 scope.
- **Downstream metric impact.** CEH's small AUC lift is +0.0015 over post-patch softmax, within the noise band of the 9-way comparison. We do not claim it as a significant improvement.

#### 4.9.4 Attribution Quality Evaluation (Post-hoc)

Because the MV alone only shows the head **trains**, we ran a dedicated post-hoc evaluation on 5,000 validation samples to ask whether the trained head produces meaningful per-sample explanations or merely a smoothed global pattern. Four measurements:

| Measurement | Value | Interpretation |
|---|---|---|
| Spearman corr. (CEH vs. grad × input) | mean 0.259, median 0.252 | Partial fit to training target |
| Between-sample / within-sample variance | 0.055 | Attribution varies much more within a sample than across samples |
| Mean top-10 feature overlap across samples | 0.791 | Different samples share ∼ 80% of their top features |
| Stability under input noise ($\sigma = 0.05$) | Spearman 0.985 | Very stable (trivially consistent with near-global output) |

Table 17: CEH attribution quality measurements on the post-patch CEH checkpoint. Taken together, the low between-over-within ratio (0.055) and high top-10 overlap (0.791) indicate the head has largely collapsed to a global importance vector with only small per-sample perturbations — the opposite of what a per-prediction attribution head should produce.

Per-group attribution mass corroborates the flatness. CEH allocates 32.1% to txn_behavior versus 44.7% under grad × input on the same samples, and overweights product_holdings (12.6% vs. 5.5%) and product_hierarchy (11.3% vs. 3.8%). The learned distribution is demonstrably flatter than its own training target.

**Interpretation.** The head fits its target only weakly ($\rho \approx 0.26$) and discards the per-sample component of that target almost entirely. The most likely mechanism: the target itself — grad × input of the causal encoder's summed output — has a large sample-invariant component, and a 64-hidden single-layer MLP can capture that global component with low loss while ignoring the sample-specific residual. The head is not broken; the target is too flat for a thin MLP to be forced into per-sample discrimination.

**What this rules in/out.**

- Rules in: infrastructure path (Finding 9 MV) is correct — a functional DAG feeds an attribution consumer without destabilising primary prediction or the DAG itself.
- Rules out: the current target design (grad × input of causal-encoder output sum) is **not** sufficient to produce a regulator-usable per-sample explanation. Additional Axis-3 candidates that depend on per-sample attribution quality (CRCG in particular) should not be evaluated against this baseline without target refinement.

**Target refinement candidates.**

1. **Demeaned target:** subtract the batch mean of grad × input before using it as supervision. Forces the head to learn per-sample deviation from the global pattern rather than re-learning the pattern itself. Smallest code change; tests the "target is too flat" hypothesis directly. **Run; see Section 4.9.4 below.**
2. **Primary-prediction target:** replace grad of the causal encoder output with grad of a specific task logit (e.g., churn_signal). Aligns the attribution with what downstream consumers actually ask explanations for, at the cost of per-task heads.
3. **Larger / deeper head:** doubled hidden dim and an extra layer, to test whether the 64-hidden bottleneck is the limiting factor rather than the target.
4. **DAG-path target:** use the learned $\boldsymbol{W}$ to construct feature-to-output influence paths as supervision, shifting from local gradient-based to structural-graph-based attribution.

#### 4.9.5 Iteration v2: Demeaned Target Resolves the Collapse

We ran candidate 1 (demeaned target) as a minimal-intervention test of the "target is too flat" hypothesis. Config flag `ceh.target_mode: "demeaned"` subtracts the per-feature batch mean from the grad × input supervision before MSE; everything else identical to the v1 MV run (same architecture, same hyperparameters, same data, same 10 epochs on SageMaker g4dn.xlarge). The same post-hoc evaluation on 5,000 validation samples produces:

| Measurement | Raw (v1) | De-meaned (v2) | Change |
|---|---|---|---|
| Between-sample / within-sample variance | 0.055 | 0.719 | 13 × larger |
| Top-10 feature overlap across samples | 0.791 | 0.281 | 65% smaller |
| Stability under input noise ($\sigma = 0.05$) | Spearman 0.985 | Spearman 0.953 | Still stable |
| Primary AUC (churn_signal) | 0.6866 | 0.6870 | Within noise |

Table 18: CEH attribution quality, v1 (raw grad × input target) vs. v2 (demeaned target). The two discriminative-power measurements move by more than an order of magnitude while primary AUC is unchanged — the head now learns per-sample deviation rather than reproducing a global importance vector, at no cost to the downstream task.

The result confirms the hypothesis from Section 4.9.3: the collapse was a training-target artefact, not a head-capacity or architectural limitation. Minimum intervention (three lines plus a config flag) restores per-sample discrimination.

Per-group attribution mass also rebalances substantially under the demeaned target. txn_behavior drops from 32.1% to 18.3% and gmm_clustering from 28.5% to 21.4%, while product_hierarchy jumps from 11.3% to 30.3% and product_holdings from 12.6% to 21.3%. The demeaned head prefers feature groups that carry per-sample distinguishing signal (product taxonomy) over the globally high-variance groups that dominated the raw head.

**Caveats.**

- Spearman correlation against **raw** grad × input falls from 0.259 to 0.096 between v1 and v2. This is expected and not a regression: the v2 head is supervised against demeaned grad × input, so the raw version is no longer its training target. Quality is measured by the target-independent metrics in Table 18 (variance ratio, top-K overlap, stability).
- Per-sample discrimination is necessary but not sufficient for regulator-usable explanations. Human evaluation on sample cases, alignment with domain expectations, and audit-log integration remain outside this paper's scope.
- We report v2 as a single-seed 10-epoch run on Santander. Cross-seed stability and cross-dataset reproduction are not verified here.

**Next iteration directions (not run).** Primary-task-gradient target remains the natural next test: replacing the causal-encoder output sum with a specific task logit aligns the attribution with what downstream consumers actually ask about ("why did this customer receive a high churn score?"). The demeaned infrastructure stays, only the scalar the gradient is taken against changes. Larger head and DAG-path targets remain as lower-priority alternatives.

#### 4.9.6 Why CEH First (Not CG / CTGR / CRCG / CCP)

Of the five Axis-3 candidates, CEH has the smallest structural footprint: a single MLP head with an MSE supervision signal, no change to task routing (vs. CTGR), no serving-time path branching (vs. CG), no cross-module wiring (vs. CRCG tying into Paper 2′s reason generator). This lets us confirm the basic premise — that downstream consumers can extract something meaningful from the now- functional DAG — with the cleanest possible test before committing to a heavier redesign. Follow-up candidates are evaluated against CEH's baseline rather than against no-causal-routing-at-all.

### 4.10 Finding 10: Causal Guardrail (CG) — Second Axis-3 Attempt

Where CEH answers "why did the model recommend this?", CG (Causal Guardrail) answers the adjacent question "can we trust this recommendation?" — a per-prediction reliability signal that routes suspicious inputs to a fallback or to human review rather than silently passing them through. SR 11-7 "known limitations" reporting and EU AI Act Art. 9 risk-management both require a runtime mechanism for this; the natural hook on the causal expert is its DAG structure.

#### 4.10.1 MV Formulation v1 (W-Reconstruction) Fails

The first formulation reused the NOTEARS reconstruction residual from Finding 8. Training optimises $\|\boldsymbol{z} - \boldsymbol{z}\boldsymbol{W}^2\|_F^2$ to keep $\boldsymbol{W}$ out of the zero saddle; at serving time the per-sample version $\|\boldsymbol{z}_i - \boldsymbol{z}_i\boldsymbol{W}^2\|^2$ is a direct measure of how well the **learned DAG** reconstructs the input's compressed causal state. A well-fit input reconstructs cleanly; an OOD input should not.

We evaluated on the teacher_ceh_demeaned checkpoint with 5,000 validation samples plus three synthetic OOD probes (uniform random in [−3,3], column-permuted, extreme-tail percentiles). All distributions concentrated within a narrow window near 1.0 (in-dist median 0.9995, p99 1.0055; OOD medians 0.9998–0.9972). At the p95 threshold calibrated on the in-distribution set, TPR on OOD probes was 6.8% / 8.1% / 0% — effectively chance-level versus the 5% FPR baseline.

The failure mechanism is the same "decorative DAG" observation from Finding 8: the learned $\boldsymbol{W}$ is small enough ($\|\boldsymbol{W}\|_F \approx 0.36$ on an $8 \times 8$ matrix, so $\boldsymbol{W}^2$ is $O(0.13)$ in Frobenius) that $\boldsymbol{z}\boldsymbol{W}^2$ barely perturbs $\boldsymbol{z}$. The residual ratio $\approx 1.0$ regardless of whether the input fits the DAG, so the signal is degenerate.

#### 4.10.2 MV Formulation v2 (z-Space Mahalanobis) Works

Since $\boldsymbol{W}$ itself is too weak to drive discrimination, we tested whether the **causal latent** $\boldsymbol{z}$ carries distributional signal independently. Let $\boldsymbol{\mu}, \boldsymbol{\sigma}$ be the per-dimension mean and standard deviation of $\boldsymbol{z}$ over the in-distribution batch; the score is the standardised-sum-of-squares $d_i = \sum_j \left(\frac{z_{i,j}-\mu_j}{\sigma_j}\right)^2$.

| Probe | median | p95 | p99 | max |
|---|---|---|---|---|
| In-distribution | 23.6 | 62.8 | 268.1 | 983.6 |
| Uniform random | 749.5 | 1200.9 | 1458.5 | 1936.7 |
| Column-permuted | 537.4 | 1076.0 | 1336.1 | 1873.9 |
| Extreme-tail | 479.3 | 479.3 | 479.3 | 479.3 |

Table 19: z-space Mahalanobis CG v2 score distributions on 5,000 validation samples and three synthetic OOD probes. Every OOD median lies above the in-distribution p99, giving near-perfect separation.

| Probe | TPR @ ID p95 | FPR @ ID p95 |
|---|---|---|
| Uniform random | 100.0% | 5.0% |
| Column-permuted | 100.0% | 5.0% |
| Extreme-tail | 100.0% | 5.0% |

Table 20: OOD detection rates at the recommended CG v2 threshold (in-distribution p95 = 62.8). All three probe types flag at the expected false-positive rate.

CG is therefore operational via the causal latent even though it fails via the DAG weights. The latent carries structure the W-reconstruction does not expose. Serving integration: a caller computes $\boldsymbol{\mu}, \boldsymbol{\sigma}$ once from a reference population, caches them, and at inference time thresholds the per-sample score to decide pass-through vs. fallback.

#### 4.10.3 What CG MV Verifies / Does Not

Verified:

1. The z-space formulation detects three qualitatively different OOD probes at effectively 100% TPR with 5% FPR — the primitive works.
2. v1 vs v2 comparison isolates **where** the signal actually lives: the causal latent, not the learned DAG weights. Reinforces the Finding-8 observation that $\boldsymbol{W}$ is structurally present but under-utilised by the current architecture.
3. Zero training cost: CG is a post-hoc analysis on the existing teacher_ceh_demeaned checkpoint. No new SageMaker job.

Not yet verified:

- **Real-world OOD**: the three probes are synthetic. A production guardrail also needs to fire on realistic distribution drift (temporal shift, subgroup imbalance, adversarial perturbation). These are Paper 2 monitoring responsibilities, not tested here.
- **Downstream metric impact**: CG produces a flag; we have not yet measured whether routing flagged predictions to a fallback improves end-to-end task metrics or calibration. Demands an integration with the 3-layer fallback router.
- **Threshold drift**: $\boldsymbol{\mu}, \boldsymbol{\sigma}$ computed on a reference batch will need periodic recalibration as the input distribution evolves. Not tested here.

#### 4.10.4 Implications for Remaining Axis-3 Candidates

The v1/v2 contrast clarifies what to expect for CTGR and CRCG. Both depend on the **learned causal structure** ($\boldsymbol{W}$) being informative in the current architecture, not merely present. The Finding 10 experiment is direct evidence that $\boldsymbol{W}$ is not currently strong enough for structural uses: CTGR's routing and CRCG's reason paths would inherit the same weak signal that doomed CG v1. Either W needs to be amplified (larger init, stronger recon lambda, or DAG-routed residual path) before those candidates are worth running, or CTGR/CRCG should be redesigned to draw from the latent rather than the weights — mirroring the v1→v2 pivot here. Finding 11 (below) runs the amplification experiment and produces a partially mixed answer.

### 4.11 Finding 11: W-Amplification Test

Finding 10 identified two hypotheses for why CG v1 fails: (a) the learned $\boldsymbol{W}$ is simply too small to

drive meaningful reconstruction, or (b) the $\|\boldsymbol{z} - \boldsymbol{z}\boldsymbol{W}^2\|^2$ formulation is structurally limited regardless of $\boldsymbol{W}$'s magnitude. We tested (a) by re-training one scenario (teacher_ceh_w_amp) with two amplification knobs: $\boldsymbol{W}$ init scale $0.1 \to 0.3$ and $\lambda_{\text{recon}}$ $0.5 \to 2.0$. All other hyperparameters identical to teacher_ceh_demeaned.

#### 4.11.1 Training-Side Result: W Amplifies, Primary Unharmed

| Metric | Baseline | W-amp | Change |
|---|---|---|---|
| $\|\boldsymbol{W}\|_F$ | 0.363 | 5.028 | 13.9 × |
| Active edges ($\|W\| > 0.01$) | 8.5% | 59.5% | 7.0 × |
| Max $\|W_{ij}\|$ | 0.11 | 0.77 | 7.0 × |
| Primary AUC (churn_signal) | 0.6870 | 0.6865 | within noise |
| Loss | 25.62 | 25.61 | within noise |

Table 21: W-amplification training-side outcome. The learned adjacency matrix grows by an order of magnitude in every structural measure while primary task metrics are preserved. Finding 8′s "decorative DAG" observation is directly and cheaply reversed.

The amplification is free in task-metric terms: the larger $\boldsymbol{W}$ pushes $\boldsymbol{W}^2$ from Frobenius 0.13 to 2.19, so the SCM residual $\boldsymbol{z} + \boldsymbol{z}\boldsymbol{W}^2$ now contributes a meaningfully non-trivial perturbation, yet primary AUC shifts by only 0.0005 (within noise). The decorative-DAG regime was a training choice (too-small init + too-weak recon term), not an architectural constraint.

#### 4.11.2 CG v1 Improves but Remains Architecturally Limited

With a 14× larger $\boldsymbol{W}$, CG v1′s discriminative power does increase but does not approach v2′s ceiling:

| Probe | Baseline TPR @ ID p95 | W-amp TPR @ ID p95 |
|---|---|---|
| Uniform random | 6.8% | 22.7% |
| Column-permuted | 8.1% | 18.6% |
| Extreme-tail | 0.0% | 0.0% |

Table 22: CG v1 OOD detection before and after W-amplification (FPR fixed at 5%). Discrimination improves from chance to weak-but-nonzero for two probe types; extreme-tail stays at zero regardless, exposing a structural limit.

The extreme-tail result is the key signal. That probe sets every sample to the same 99th-percentile vector, so every sample has an identical $\boldsymbol{z}$ and therefore an identical residual — $\|\boldsymbol{z} - \boldsymbol{z}\boldsymbol{W}^2\|$ is a point estimate, unable to discriminate by construction. No amount of W amplification fixes this because the formulation is a **per-sample** measure that offers no vantage point on the **distribution** of inputs. CG v2′s z-space Mahalanobis, by contrast, is explicitly a distance from a reference distribution, so it handles the degenerate-OOD case trivially (100% TPR).

#### 4.11.3 CG v2 Unaffected

v2′s discrimination is unchanged after amplification: 100% TPR across the three probes at the recommended p95 threshold. The z-space pathway does not depend on $\boldsymbol{W}$'s magnitude; amplification neither helps nor hurts it.

#### 4.11.4 What Finding 11 Settles / Leaves Open

Settled:

1. $\boldsymbol{W}$ is **not** architecturally doomed. With init 0.3 + $\lambda_{\text{recon}} = 2.0$, it grows to a non-trivial sparse structure in 10 epochs, with zero primary-task cost.
2. CG v1′s failure in Finding 10 was **partly** due to small $\boldsymbol{W}$ (now contributes 15–20 TPR points) and **partly** due to a formulation ceiling (extreme-tail probe exposes a structural limit).
3. Latent-space CG (v2) remains dominant for OOD detection; W-based guardrails are a supplement at best, not a replacement.

Open:

- Does amplified $\boldsymbol{W}$ unlock CTGR and CRCG, or do those candidates hit similar structural ceilings? Finding 11 answers only the CG question directly; the training-side amplification means CTGR/CRCG can now at least be **tried** without the Finding 10 preconditions blocking them.
- Does an even more aggressive amplification (init 0.5, $\lambda_{\text{recon}} = 5$, removed sparsity) push v1 to a usable level or crash primary AUC? Not tested; the $\frac{0.3}{2.0}$ setting was chosen as a single-job minimum-intervention.
- Does latent-space CG's 100% TPR survive **realistic** drift (temporal, subgroup, adversarial) rather than synthetic probes? Paper 2 monitoring scope, not tested here.

#### 4.11.5 Updated Recommendation for Remaining Axis-3 Candidates

Finding 10 recommended either amplifying $\boldsymbol{W}$ or redesigning the siblings around the latent. Finding 11 chooses the first option and shows it partially works: $\boldsymbol{W}$ is now meaningful, but the latent-based formulation still ceiling-beats a W-based one when both are available. The practical recommendation for CTGR / CRCG / CCP is therefore:

1. Run them on a W-amplified teacher, so W is **capable** of carrying structural information.
2. Simultaneously evaluate a latent-based alternative for each candidate. In every case where both are measurable, treat the latent version as the baseline to beat rather than assuming the W-based version is preferable.

# 5 Discussion

## 5.1 Practical Guidelines Summary

We distill six findings into five guidelines for practitioners:

1. **Gate selection depends on task-type mix, not on architecture.** Use softmax for heterogeneous task mixes (different loss types); use sigmoid for homogeneous tasks (all binary, or all regression). This is the single highest-impact design decision.
2. **Uncertainty weighting is necessary but not sufficient.** It normalizes loss scales — without it, multiclass tasks are silently suppressed. But it does not prevent gradient contamination. Gate structure provides the actual isolation. Verify the implementation against Equation 3, especially the per-task $w_t$ and clamping.
3. **Do not mix task types in pre-gating groups.** GTE, task-group attention, and similar mechanisms assume intra-group homogeneity. Business-meaningful groups that mix binary and regression tasks will degrade the minority type.
4. **Use a composite checkpoint metric, not val loss.** When regression and classification tasks share a composite loss, regression improvement continuously pulls val loss downward while classification/ranking metrics saturate or regress. Define a type-weighted composite metric (Avg AUC + NDCG@3 + normalized MAE) before training and checkpoint by it (Section 4.6).
5. **Gate entropy reveals architectural waste.** If all tasks show uniform attention-level entropy (ratio = 1.000), the attention aggregation is not performing routing — it is averaging. Audit entropy ratios at both the extraction and attention levels before attributing performance gains to gating mechanisms.

## 5.2 Limitations

**Synthetic data**: All experiments use a synthetic benchmark with controlled noise profiles. Real production data may exhibit different gradient dynamics due to label sparsity, class imbalance, and non-stationary distributions. We plan to supplement with production results as they become available.

**Single expert basket**: Our findings are specific to PLE with 7 heterogeneous experts. Homogeneous-expert PLE may exhibit different gate dynamics.

**Epoch budget and task-type interaction**: Finding 4 acknowledges that 10-epoch comparisons may be premature. Finding 6 extends this to 30 epochs and confirms that additional epochs help regression but harm classification and ranking. Cross-architecture comparisons (e.g., PLE vs.
shared-bottom) at 30 epochs remain pending.

**Single dataset scale**: While 1M customers is representative of mid-sized financial institutions, the findings may not generalize to internet-scale datasets (100M+ users) where task gradient dynamics differ.

## 5.3 Relationship to Companion Papers

This paper complements two companion papers from the same project: **Paper 1** (architecture and ablation) establishes the heterogeneous expert PLE design and validates expert specialization via joint feature+expert ablation. **Paper 2** (serving and ops) covers knowledge distillation, recommendation reason generation, and regulatory compliance. The present paper focuses specifically on **loss dynamics and gating behavior** that emerged during the ablation study but warranted deeper analysis than Paper 1′s scope allowed.

# 6 Conclusion

Scaling multi-task learning from 2–4 homogeneous tasks to 13 heterogeneous tasks reveals dynamics that existing literature, evaluated primarily on homogeneous setups, does not address. The gate type choice — softmax vs. sigmoid — depends not on architectural preference but on whether tasks share the same loss type. Uncertainty weighting normalizes scales but does not isolate gradients. Pre-gating mechanisms like GTE require type-homogeneous groups. Training budgets must account for cosine restart cycles before structural comparisons are meaningful.

Gate entropy analysis adds further precision to this picture: PLE extraction-layer gating demonstrably specializes (entropy ratios 0.33–0.88), with simple-pattern tasks concentrating on 1–2 experts and complex tasks distributing weight across all 7. Attention-level aggregation, however, collapses to uniform averaging (entropy ratio 1.00 for all tasks), suggesting that this component does not perform routing in practice. Finally, composite val loss is an unreliable checkpoint signal once regression tasks are present — their continuous improvement masks

classification and ranking metric degradation, and cosine learning-rate restarts amplify this divergence by disproportionately disrupting the ranking-optimal parameter region.

These findings are not novel algorithms but practical diagnostics. We hope they prevent other practitioners from re-discovering the same pitfalls when scaling MTL to real-world heterogeneous task portfolios.

Finding 7 maps the search space for fusion augmentation on top of CGC into three regions. **Representation-additive fusions** — loss-level adaTT, AdaTT-sp, M1 complement, ECEB, and MV BRP — all propagate residual-error signal into the primary-representation path and uniformly degrade aggregate AUC, with magnitude scaling monotonically in the invasiveness of the intervention. **Output-space boosting with shared-expert gradient isolation** (BRP-detached) ties CGC on aggregate AUC ($\Delta = -0.0007$, best epoch $0.6736 > 0.6728$) while improving F1 macro by $+0.007$ and NDCG@3 by $+0.015$ and retaining a $+256\%$ relative rescue on the hardest multiclass task. **Training-time load-balancing regularisation** (NEAS) is the first mechanism of the family to actually raise aggregate AUC ($\Delta = +0.0011$), with a monotone-increasing trajectory and near- uniform per-task lifts. The two positive recipes act on disjoint axes (error correction in output space vs. expert-collapse prevention at the gate) and are **not additive**: stacking them collapses NEAS's AUC gain because the shared experts cannot simultaneously be generalists (NEAS) and primary-supporting specialists (BRP-detached). The practical guidance is per-objective: NEAS for aggregate AUC and uniform cross-task robustness, BRP-detached for hard-task rescue. Follow-up work will validate both recipes across seeds and datasets and evaluate mechanisms that could resolve the NEAS $\times$ BRP- detached conflict (phased-schedule training, parameter-shared heads) in a SageMaker setting where the compute budget allows systematic sweeps.

# 7 Author Contributions

**Seonkyu Jeong** (PM / Lead Architect / Data Scientist): Conceived the study, designed the ablation framework, identified all five findings, wrote the manuscript. Led AI-augmented development methodology.

**Euncheol Sim**: Data pipeline, feature engineering, ablation execution.

**Youngchan Kim**: Model training, mathematical verification, loss weighting implementation.

All authors collaborated through Scrum sprints with rapid feedback cycles.


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